

Physics 11 Practice:

Power + Efficiency + Mechanical Advantage

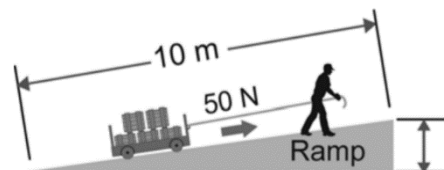
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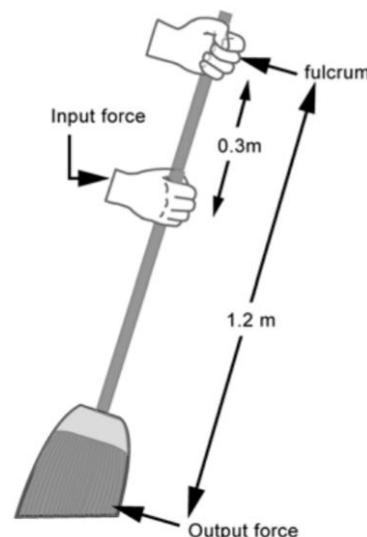
1. Find work done by a bulldozer pushing a rock with a force of 5000 N at 2.0 m/s for 20 s.
2.
 - a) How much work (in kW·hrs) is done by a 6000 W electric generator running 8 hours a day for an entire year?
 - b) How long (in hours) will it take a 500 W power drill to do 100 kW·hrs of work?
 - c) How much power is being developed by a machine that can do 600 kW·hrs of work in 12 hr?
3.
 - a) Find power generated by a 70.0 kg person running up a 3.00 m high flight of stairs in 2.20 s
 - b) Find time required for person generating same power as above to run up a 4.5 m flight of stairs.
4.
 - a) Find time required for a 5.0 hp motor to lift a 500 kg safe up to a window 30 m above ground?
 - b) Find power required to move a 100.0 kg box a distance of 4.0 m along a level surface ($\mu = 0.25$) with constant velocity over a period of 3.6 s.
5. A farmer did 1.38 kJ of work in pushing a 240 kg cart a certain displacement from rest. After farmer had pushed cart a certain displacement, it was moving with a velocity of 2.2 m/s.
 - a) Find cart's change in kinetic energy
 - b) Find efficiency of this process
6. A worker performed a one-time lift of a 2.0 kg mass (initially at rest) a vertical distance of 10.0 m from the ground using a rope, achieving a final speed of 7.3 m/s. When a motor was used to perform this same task repeatedly, the motor achieved a final speed of only 5.5 m/s at 10.0 m. How efficient was the motor?
7. On a stair climber at a gym, a 60.0 kg person lifts their centre of mass 20.0 cm with each step. If they take 3.0×10^2 steps in 6.0 minutes and their muscles are 20% efficient, find:
 - a) rate at which person does work
 - b) rate at which person uses food energy
8.
 - a) A kettle is 82% efficient and has a power rating of 1200W. Find rate that water in kettle absorbs energy.
 - b) A light bulb brand has an efficiency of 4.8%. For a 60W rated bulb of this brand, find rate at which bulb produces light energy
 - c) For every MJ (1×10^6) of chemical potential energy in vehicle fuel, only 125 kJ of useful work is done by vehicle making itself move.
 - i. How efficient is vehicle?
 - ii. Where are some places that energy from fuel is wasted?

9. a) 620 J of work is done to operate a pulley system that lifts a 160N load to a height of 3.1m.
Find efficiency of pulley system.
- b) Find work done by a person who uses a pulley that is 90.3% efficient to raise a 52.0 kg mass to a height of 61.3 cm.

10. To lift a 220 N cart vertical height of 1.8 meters, a person pulls the cart up a 10.0 m ramp with an effort force of 50.0 N.
Find: a) actual mechanical advantage of scenario
b) efficiency of ramp



11. A person pulls a rope attached to a pulley system a distance of 3.90 m, exerting a force of 375 N, causing pulley system to lift a 1345 N weight a distance of 0.975 m.
a) find actual mechanical advantage of scenario
b) find efficiency of pulley system

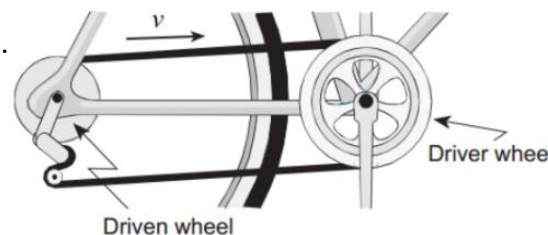


12. A person was sweeping their floor as shown (lever system) with an input (effort) force of 36 N.
If their sweeping action was 42% efficient, find output force would be produced (sweeping a load).

13. a) A pedal, gear & rear axle system on a bicycle produces 430 N at gear for every 1.00×10^3 N applied to pedals.
System = 23% efficient.

<https://www.sportsperformancebulletin.com/training/techniques/can-your-peddaling-technique-make-you-a-more-efficient-rider>
If gear/rear axle setup has a diameter of 26 cm, find diameter of pedal/chain setup.

- b) For the bicycle pedal system shown, driver (larger) wheel has 32 teeth and radius = 12 cm.
In this system, mechanical advantage = $1/3.5$.
Find number of teeth vs. radius for driven (smaller) wheel.



KEY

1. 200 kJ
2. a) 20000 kWh b) 200 h c) 50 kW
3. a) 935 W b) 3.3 s
4. a) 39 s b) 270 W
5. a) 580 J b) 42%
6. 91%
7. a) 98 W b) 490 W
8. a) 980 W b) 2.9 W c) i. 12.5% ii. heat from friction between moving parts
9. a) 80 % b) 346 J
10. a) 4.4 b) 79%
11. a) 3.59 b) 89.7%
12. 3.8 N
13. a) 49 cm b) 9 teeth vs. 3.4 cm